DOKUMENTASI JARINGAN

Konfigurasi Internet 100 Mbps Single

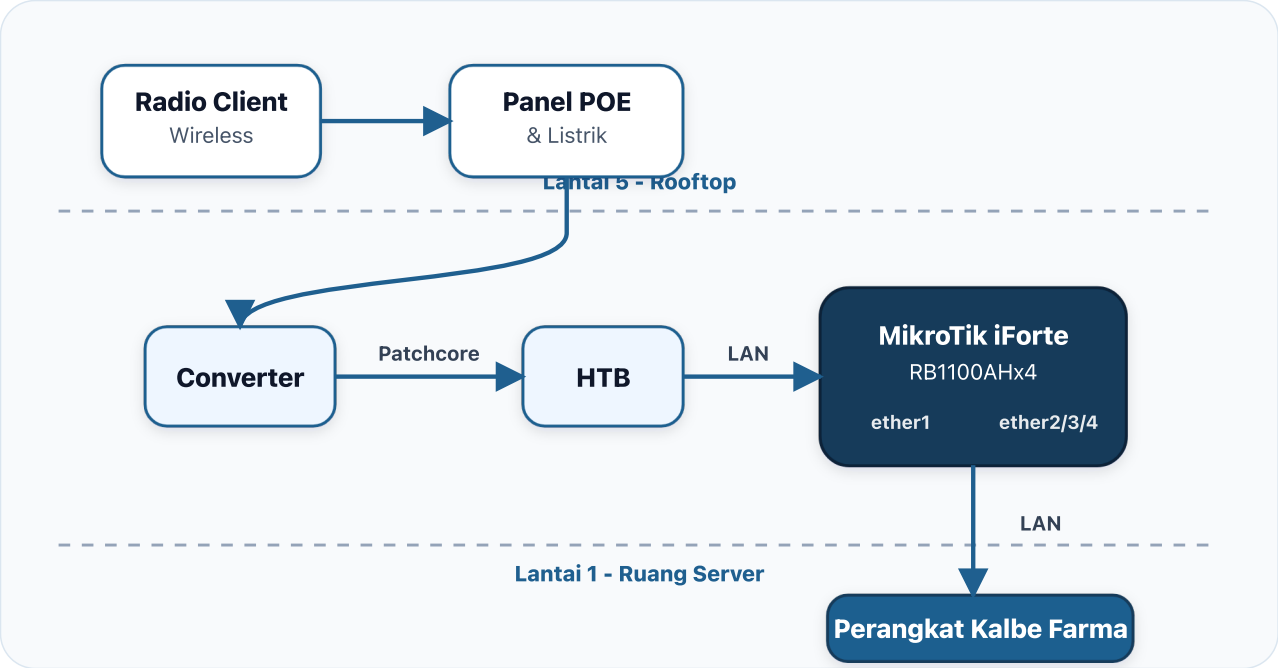
SI069764 - Kalbe Farma Cikarang

CUSTOMER **Kalbe Farma Cikarang**	SERVICE ID **SI069764**
ROUTER **MikroTik iForte RB1100AHx4**	TIME ZONE **Asia/Jakarta**

1. Informasi Layanan

Item	Keterangan
Site / Customer	Kalbe Farma Cikarang
Service ID	SI069764
Layanan	Internet 100 Mbps Single
Router	MikroTik iForte RB1100AHx4
Lokasi Radio / POE	Lantai 5 - Rooftop
Lokasi Router / Server	Lantai 1 - Ruang Server
Time Zone	Asia/Jakarta

2. Topologi Jaringan



Alur koneksi dimulai dari Radio Client di rooftop, turun melalui Panel POE, Converter, HTB, lalu masuk ke MikroTik iForte RB1100AHx4 di ruang server sebelum diteruskan ke perangkat Kalbe Farma.

Perangkat	Fungsi	Lokasi
Radio Client	Menerima koneksi wireless dari provider	Lantai 5 / Rooftop
Panel POE & Listrik	Menyediakan power ke radio client	Lantai 5 / Rooftop
Converter	Konversi media koneksi dari sisi rooftop	Jalur menuju ruang server
HTB	Terminasi koneksi sebelum masuk router	Ruang Server
MikroTik iForte RB1100AHx4	Router utama untuk WAN, Public IP, LAN, NAT, DHCP, DNS, SNMP	Lantai 1 / Ruang Server
Perangkat Kalbe Farma	Perangkat LAN/customer yang menerima koneksi internet	Lantai 1 / Ruang Server

3. Mapping Interface MikroTik

Interface	Fungsi	Keterangan
`ether1`	WAN Wireless	Interface fisik untuk VLAN WAN
`WAN Wireless`	VLAN WAN	VLAN ID `2455` di atas `ether1`
`ether2`	Public IP LAN	Digunakan untuk IP publik customer
`ether3`	LAN Private	Masuk ke bridge private
`ether4`	LAN Private	Masuk ke bridge private
`Bridge-LAN_Private`	Bridge LAN private	Bridge untuk jaringan private `192.168.100.0/24`

4. Konfigurasi VLAN WAN

Parameter	Nilai
Name	`WAN Wireless`
Interface Parent	`ether1`
VLAN ID	`2455`

VLAN 2455 dibuat di atas ether1 dan digunakan sebagai jalur WAN wireless dari provider menuju router MikroTik.

5. IP Addressing

IP WAN	Nilai
Address	`100.70.161.142/30`
Interface	`WAN Wireless`
Gateway WAN	`100.70.161.141`

IP Public Customer	Nilai
Address	`103.165.123.185/29`
Interface	`ether2`
Subnet Mask	`255.255.255.248`
IP Available	`103.165.123.186 – 103.165.123.190`
Gateway Customer	`103.165.123.185`

IP LAN Private	Nilai
Address	`192.168.100.1/24`
Interface	`Bridge-LAN_Private`
Network	`192.168.100.0/24`

Catatan: karena ether3 dimasukkan ke Bridge-LAN_Private, IP LAN private idealnya dipasang pada interface bridge.

6. Routing, DNS, dan NAT

Static Route	Nilai
Dst. Address	`0.0.0.0/0`
Gateway	`100.70.161.141`

DNS	Nilai
Primary DNS	`202.51.96.7`
Secondary DNS	`202.51.102.118`
Allow Remote Request	Disabled

NAT Firewall	Nilai
Chain	`srcnat`
Src. Address	`192.168.100.0/24`
Action	`src-nat`
To Address	`103.165.123.185`

Trafik LAN private 192.168.100.0/24 keluar internet menggunakan NAT ke IP publik 103.165.123.185.

7. Bridge LAN Private

Bridge	Nilai
Name	`Bridge-LAN_Private`
STP	`none`

Interface	Bridge
`ether3`	`Bridge-LAN_Private`
`ether4`	`Bridge-LAN_Private`

8. IP Service

Service	Port	Status
Telnet	`27292`	Enabled
Winbox	`36459`	Enabled

Service	Status
FTP	Disabled
SSH	Disabled
WWW	Disabled
API	Disabled
API-SSL	Disabled

9. DHCP Server

Parameter	Nilai
DHCP Server Interface	`Bridge-LAN_Private`
DHCP Address Space	`192.168.100.0/24`
Gateway for DHCP Network	`192.168.100.1`
Address to Give Out	`192.168.100.2 – 192.168.100.250`
DNS Server	`202.51.96.7`, `202.51.102.118`
Lease Time	Default / sesuai kebutuhan

10. SNMP

Parameter	Nilai
SNMP	Enabled
Community Name	`durentiga`
Community Address	`0.0.0.0/0`
Trap Community	`durentiga`
Trap Version	`2`

11. System Identity & Time

Parameter	Nilai
Identity	`SI069764_INTERNET_100M_SINGLE_KALBE_FARMA_CIKARANG`
Time	Disesuaikan
Date	Disesuaikan
Time Zone Name	`Asia/Jakarta`

12. Ringkasan IP Publik

Item	Nilai
Network	`103.165.123.184/29`
Gateway	`103.165.123.185`
Usable IP	`103.165.123.186 – 103.165.123.190`
Broadcast	`103.165.123.191`
Subnet Mask	`255.255.255.248`
DNS 1	`202.51.96.7`
DNS 2	`202.51.102.118`
Customer Port	`ether2`

13. Ringkasan Alur Koneksi

1. Koneksi provider diterima oleh Radio Client di rooftop.
2. Radio mendapat power dari Panel POE & Listrik.
3. Jalur turun menuju Converter.
4. Dari Converter masuk ke HTB melalui patchcore.
5. Dari HTB masuk ke MikroTik iForte RB1100AHx4.
6. MikroTik menerima WAN melalui VLAN 2455 pada ether1.
7. Public IP customer tersedia pada ether2.
8. LAN private berjalan pada Bridge-LAN_Private melalui ether3 dan ether4.
9. Trafik LAN private keluar internet menggunakan NAT ke 103.165.123.185.